Comparing the performance of BERTje and BERT on the Dutch Political Election Manifestos

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Abstract

This study compares multiple approaches for multi-label classification of Dutch election manifestos, including Linear Support Vector Machines (SVM) with TF-IDF, the Dutch transformer model BERTje, and English transformer models (BERT Base and Large BERT) applied to translated texts using OPUS and DeepL models. The goal was to assess the impact of translation and model scale on performance.

Key findings show that the free-of-use OPUS-MT translation model performed nearly identically to the commercial DeepL model on BERT, suggesting the validation to use of OPUS-MT as a translation model. While the language-specific BERTje slightly outperformed BERT Base (both ~ 0.1B parameters) on the full 380-label set, model scale emerged as the dominant factor. Large BERT (~ 0.3B parameters) on translated English text achieved a significantly higher F1 score (0.47 on all three label years) than all other models, confirming that increasing model capacity provides substantial gains that outweigh the benefits of smaller, language-specific models for this task. The transformer models also achieved a better balance between precision and recall compared to the SVM baseline.

1 Introduction

Dutch-language media is considerably scarcer than English-language media. When classifying Dutch election manifestos, Dutch transformer models can be used, but their performance may be limited by the relatively small amount of Dutch training data. An alternative approach is to translate the manifestos into English and apply larger, better-trained English transformer models. This study compares multiple approaches for multi-label classification of Dutch election manifestos: Linear Support Vector Machines (SVM) with TF-IDF, Dutch transformer model, and English transformer model applied to translated texts. The following research questions are addressed:

1. How does model based translation of the dataset impact performance of BERT on the dataset?
2. How does BERTje compare to BERT in a multi-label classification task?
3. How much does large BERT improve over base BERT on the translated dataset?

RQ1: Translation may introduce information loss or distort Dutch-specific context. However, English transformer models benefit from much larger training corpora, which may offset this drawback. Investigating if different translation models have a large impact on performance on this classification task is crucial to understand the impact of information loss. The impact of translation is evaluated for both transformer-based models and Linear SVM, which is expected to be more sensitive to translation errors.

RQ2: Comparing these models in a multi-label classification setting reveals whether language-specific pretraining outweighs the advantages of model scale and generalization from larger English datasets. This comparison informs whether, for tasks in a low-resource language, translation or native models yield better performance.

RQ3: Base BERT and BERTje have the same parameter size of 0.1B. BERT large has 3 times the amount of parameters and might showcase an improvement on the translated dataset compared to base BERT, we are interested in finding the amount of improvement.

2 Related work

Some important related work is briefly mentioned.

2.1 Linear SVM and TF-IDF

Linear Support Vector Machines (SVMs) combined with Term Frequency–Inverse Document Frequency (TF–IDF) representations have been the baseline for text classification tasks. Linear SVMs are particularly good in high-dimensional sparse spaces[6]. TF–IDF weighting further enhances discriminative power by down-weighting frequent but uninformative terms. Due to their scalability, robustness, and strong empirical performance, linear SVMs with TF–IDF remain a strong baseline in classification problems, even compared to more complex neural models. For this project, the sci-kit learn library is used[5].

2.2 BERT

Bidirectional Encoder Representations from Transformers (BERT) [4] is a pre-trained language model based on the Transformer encoder architecture[8]. This allows BERT to learn context that can be fine-tuned with minimal architectural changes for a wide range of downstream natural language processing tasks, including text classification. For the comparisons base BERT (0.1B parameters) and large BERT (0.3B parameters)[3] will be used.

2.3 BERTje

BERTje is a Dutch pre-trained language model based on the BERT architecture [1]. It follows the same Transformer encoder design and pre-training techniques as BERT, but is trained exclusively on Dutch text, including books, news articles, and Wikipedia. As a result, BERTje has shown strong performance across a variety of Dutch natural language processing tasks, such as text classification, named entity recognition, and part-of-speech tagging.

2.4 Dutch Election Manifestos

A common dataset for Dutch political manifestos for analysis is the Lipschits corpus, which consists of Dutch election manifestos from 1977 to 1998 that were manually annotated by the Dutch political scientist Isaac Lipschits [9]. This dataset is multi-label annotated per paragraph.

2.5 Automatic thematic classification of election manifestos

Verberne et al. [9] study the automatic thematic classification of Dutch election manifestos using the same Lipschits-annotated corpus. They frame the task as multi-label text classification and show that traditional models such as Linear SVMs and Balanced Winnow with bag-of-words features achieve precision comparable to that of a human expert, although insights into recall is limited by the way datasets are handled across years.

2.6 Translation models

To perform the translation of the Dutch dataset to English, the decision was made to primarily use a free model from Hugging Face, as utilizing the DeepL API beyond 500k characters would be expensive. We namely utilized the `opus-mt-nl-en` model [7], which is part of the OPUS-MT project and has a relatively high BLEU score of 60.9. Besides this primary model, we also used DeepL's translator [2] on the 1984 dataset to generate a small portion of the data to see if there is a difference in performance with different translations.

3 Data

The data is provided on Brightspace to us as in a common `XML` format.

3.1 Textual description

It is structured in a typical `XML`-style defined with a hierarchy of tags to store content and metadata. With the following format:

`<Lipschits>` The root element, containing a `year` attribute.

`<chapter>` Represents a political program section, identified by a `party` ID attribute.

`<p>` An individual paragraph element, containing the policy text and a unique `id`.

`<themes>` A metadata block nested within `<p>`, holding multiple `<theme>` tags.

`<theme>` Links the paragraph to specific policy topics.

It is a multi-label classification task, where each paragraph is linked to multiple themes. The goal is to predict these themes. Inside the `<p>` block the paragraphs are stored. As there are translated paragraphs. The following are snippets of the same sentence:

ORIGINAL: *Samen werken voor morgen Het CDA heeft zich in de voorbije regeerperiode metterdaad ingezet om de klemgereden samenleving in het goede spoor te brengen...*

OPUS: *Working together for tomorrow In the past ten years, the CDA has made an effective effort to bring society to the right track...*

DEEPL: *Working together for tomorrow The CDA has worked hard in the past legislature to get our society back on track...*

Before translation or performing any experiments on the dataset, the dataset had to be cleaned slightly. Some of the entries had no text, These entries were removed. For convenience, the stored data was converted to a more suitable datatype in the form `csv` files.

3.2 Statistical description

The combined dataset utilized for training encompasses 2569 examples in total, drawn from the political programs of 1986, 1994, and 1998 (as shown in Table 2). These examples are categorized by a comprehensive set of 380 unique policy themes (labels).

3.2.1 Label Sparsity Analysis

An important characteristic of the dataset is the observed label sparsity. This highlights the most consistently emphasized policy areas across the three election years. The most prevalent theme in the entire corpus is *milieubeleid* (environmental policy), which occurs in 718 samples, covering 27.95% of the dataset. Following this, *werkgelegenheid* (employment) and *Europese_samenwerking* (European cooperation) are also highly frequent, appearing in 23.86% and 18.57% of the samples, respectively.

Table 1 details the top 15 most frequent policy themes in the combined dataset, showing their absolute count and percentage coverage:

Table 1: Label Sparsity Analysis: Top 15 Policy Themes (Combined Dataset)

Theme (Dutch)	Count	Percentage (%)
milieubeleid	718	27.95
werkgelegenheid	613	23.86
Europese_samenwerking	477	18.57
gemeentelijk_bestuur_en_beleid	382	14.87
wetshandhaving	367	14.29
loon-_en_inkomensbeleid	340	13.23
sociaal_beleid	324	12.61
inrichting_van_de_staat	308	11.99
economisch_beleid	285	11.09
politiebeleid	268	10.43
ontwikkelingssamenwerking	265	10.32
minderheden_etnische_en_culturele	251	9.77
sociale_voorzieningen	246	9.58
buitenlands_beleid	241	9.38
belastingen_algemeen	231	8.99

Conversly, there are many other labels as seen in Table 2 with many unique labels only covered a few times. This means that getting a high macro F1 will prove to be difficult.

Table 2: Dataset overview per election year

Year	Training Examples	Total Labels
1986	797	214
1994	946	217
1998	826	218
1986 + 1994	1748	320
1986 + 1994 + 1998	2569	380

3.3 Challenges

The primary challenges when working with this dataset stem from the limited overall size and a significant issue of label sparsity. Addressing these factors is crucial for designing and evaluating any classification model.

1. **Limited Data Size:** Although the combined dataset offers 2,569 training examples, the data is divided across three separate election years, as detailed in Table 2. The individual year datasets are relatively small, especially considering the unique themes in each dataset.

2. **Severe Label Sparsity:** The issue of sparsity is seen by the large number of unique policy themes. With 380 total unique labels across only 2,569 paragraphs. the Label Sparsity highlights that a small fraction of themes (the top 15) is in a large portion of the data as seen in Table 1.

4 Methods

The methodology for each experiment is motivated and briefly described.

4.1 Data preperation

Using the model on Huggingface the dataset was translated to english. This was done per sentence as there was a max length limit and after translation adding all the sentences of an example together. This could potentially cause some context loss. The sentence splitting is performed by the Spacy package. A dataset of only 133 examples was created using the DeepL API. Here the translations are performed per example instead of per sentence. The cleaned original dutch text, english huggingface text and deepl text are stored in a CSV for later ease of use.

The data of all the years was combined, but tested with labels of all years or exclusively first and second or the first year.

4.2 Model settings and environment

The experiments were run on the LIACS linux workstations and the vibranium REL compute system.

4.2.1 Support Vector Machine settings

Similarly to the settings of Verberne et al. [9], the SVM with TF-IDF used the following settings:

1. min_df = 2
2. lowercase = `True`
3. ngram_range = (1,1)

There are many more options to consider such as stopwords, max features or bigger ngram ranges. However, as this is merely a baseline these were not further investigated.

4.2.2 Transformer settings

For the data clustering was used as both Bert models have a context limit of 512. The decision was made to use a max length of 384 and a stride of 128 to still have overlap between the individual chunks. At the end of evaluation the predicted labels of the same cluster are grouped together by doing a max pooling to get the final prediction of labels.

- learning rate = $2 \cdot 10^{-5}$
- batch size = 16
- epochs = 20
- weight decay = 0.01
- prediction threshold = 0.2

4.3 Evaluation

The methodology of evaluating the models is described.

4.3.1 Evaluating translation

To ensure that the proposed free translation model produces outputs of acceptable quality, we compare its performance against DeepL, a more established translation model. Due to DeepL's limitation of translating a maximum of 500,000 characters, the evaluation is restricted to the subset of the dataset that falls within this limit. Specifically, we use the 133 paragraphs from the 1984 dataset that were translated by DeepL.

For evaluation, we generate translations of this subset using both DeepL and the OPUS-based model. We then run with a 75/25 train and evaluation split to the translations produced by each model with 10 epochs. By comparing the F1 scores across the two models on the same data, we get an idea whether there is a significant difference in translation quality. This comparison serves as a sanity check to verify that the proposed free translation model performs competitively and is not inferior to a strong commercial baseline.

4.3.2 Evaluating Different Thresholds

For multi-label classification, selecting an appropriate threshold is crucial. To investigate this, we evaluate thresholds ranging from 0.1 to 0.5 in increments of 0.05. At each threshold, we measure the F1 score, precision, and recall for both the BERT and BERTje models under the various label settings, as described in the upcoming section. The SVM model, in contrast, uses the default threshold determined by the Python package. To evaluate the thresholds, a 90/10 training split is used.

4.3.3 Evaluating across Election Years

As explained in Section 3.2, the label sets differ across datasets. Training a model on a single dataset would restrict it to predicting only the labels present in that dataset, making it unable to handle unseen labels. To address this, we evaluate the model using three different label sets: 1986, 1986+1994, and 1986+1994+1998. This approach allows us to assess how well the model generalizes to each year, given that it is trained on all available years and labels. To investigate the strength of these models, the F1, precision and recall are investigated with a 90/10 training split.

4.3.4 Evaluating Large BERT

In addition to comparing English and Dutch BERT models of similar size, we also investigate the performance of a larger BERT model, which is publicly available on HuggingFace[3]. This experiment examines whether increasing model capacity provides benefits for multi-label classification in our datasets. The evaluation follows the same procedure as the other models, allowing for a direct comparison of performance and insight into the impact of model scale on classification effectiveness.

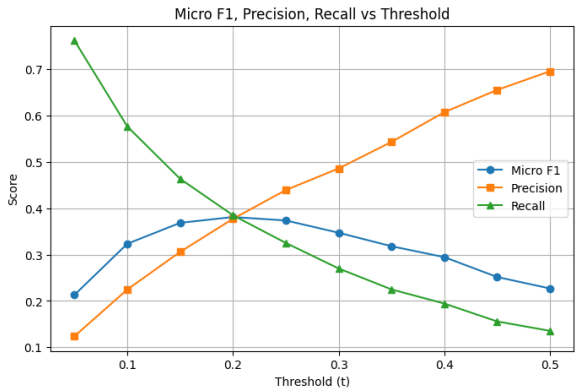

Figure 1: Visualization of F1, precision and recall across different thresholds.

5 Results

The results are reported and briefly examined.

5.1 Results across different translations

In Table 3 the comparison between the two translation models can be seen. The threshold which ended up being chosen was 0.3, because it appeared that with larger or smaller thresholds the models would both get stuck predicting the same. This could be due to the fact that the dataset was very limited at only 133 samples. However, with this small sample, the results are almost identical. This suggests that OPUS, however free-of-use, seems to be just as potent for this multi label classification task.

Table 3: Comparative BERT-based Evaluation Results on the Common Paragraph Set

Model	Micro-F1	Macro-F1	Micro-Precision	Micro-Recall	Loss
OPUS_EN	0.3436	0.0410	0.4127	0.2943	0.2982
DeepL_EN	0.3460	0.0413	0.4128	0.2979	0.3011

5.2 Results with different thresholds

Different thresholds have a significant impact on performance, as shown in Appendix A. Unlike SVMs, BERT(je) models exhibit a negative correlation between precision and recall: for each model, the optimal F1 score is achieved when precision and recall are approximately equal, as illustrated in Figure 1, where increasing one would decrease the other. This was observed for each transformer model on each label set.

5.3 Results between election years

Table 4 shows the performance of the models across different label sets. Several trends are noteworthy. First, while all models experience a decline in performance as more label years are included, Linear SVM exhibits a relatively gradual drop-off, maintaining high precision even when predicting all three label years. In contrast, BERT and BERTje experience a sharper decrease in F1 score as the number of labels grows. Interestingly, BERTje outperforms BERT when all three label years are considered, suggesting it generalizes better to larger and more diverse label sets. Another important observation is the balance between precision and recall: Linear SVM consistently achieves much higher precision than recall, whereas BERT and BERTje show more balanced precision and recall scores, in line with the discussion in the previous section. Overall, BERTje appears to have a smaller performance drop-off compared to BERT, indicating that it is more robust when handling multiple years of labels. In Table 5 the best thresholds are listed. It seems that both BERT and BERTje have a very similar threshold. The best chosen thresholds are relatively close to one another, as seen in the Appendix.

5.4 Results of Large BERT

Large BERT outperforms base BERT, BERTje, and Linear SVM, as shown in Table 4. This demonstrates that increasing model capacity is beneficial for multi-label classification on this dataset. Notably, the drop in performance is less pronounced when predicting additional labels, which suggests that larger Transformer models are better able to capture dependencies across multiple labels and maintain stronger precision and recall even as the label space expands. Furthermore, the optimal thresholds for Large BERT are generally higher than those for base BERT and BERTje, as shown in Table 5, indicating that the model is more confident in its predictions and allows for stronger decisions without sacrificing recall.

6 Discussion

Model scale emerged as the most prominent boost in performance. Large BERT on translated English text achieved an F1 score of 0.47 (Table 4), outperforming same-sized BERTje and BERT Base models as well as the Linear SVM baseline. This confirms that increasing parameters from 0.1B (Base) to 0.3B gives big perofmrance gains, with potential information loss due to translation. Free OPUS-MT translation performed almost

Table 4: Model performance per label sets.

Model	Included labels								
	1984 labels			1984+1992 labels			1984+1992+1996 labels		
	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall
Linear SVM	0.45	0.84	0.31	0.39	0.84	0.25	0.37	0.85	0.24
BERT	0.51	0.51	0.50	0.44	0.45	0.43	0.33	0.31	0.35
BERTje	0.51	0.49	0.53	0.43	0.41	0.45	0.38	0.38	0.38
Large BERT	0.57	0.57	0.56	0.49	0.53	0.47	0.47	0.52	0.43

Table 5: Best thresholds for each model and label set.

Model	First		First two		All three	
	F1	Threshold	F1	Threshold	F1	Threshold
BERTje	0.51	0.25	0.43	0.20	0.38	0.20
BERT	0.51	0.30	0.44	0.25	0.33	0.15
Large BERT	0.57	0.35	0.49	0.30	0.47	0.30

identically to commercial DeepL (Micro-F1 of 0.3436 versus 0.3460, Table 3), validating translation as a viable, cost-effective approach for this task.

When comparing models of similar capacity (roughly 0.1B), BERTje outperformed BERT Base on the full set of 380 labels, showing that Dutch pre-training better captures political language or that information is lost in translation. However, this advantage is completely overridden by the scale of Large BERT, reinforcing that model scaling is the main contributor for higher performance.

This work advances multi-label classification on the Lipschits corpus, particularly addressing the low recall identified in prior work. Linear SVM achieved high precision (0.85) but very low recall (0.24), consistent with findings from Verberne et al. [9], regardless with a slightly different experimental setup. Transformer models achieve a more balanced precision and recall as seen in Figure 1. Severe label sparsity, with many labels appearing only a few times (Table 2), made Macro-F1 negligible (Macro-F1 0.0410), justifying focus on Micro-F1, precision, and recall as metrics that better reflect performance on this label prediction task.

Performance declined as more election years and unique label sets were included in the prediction task (Table 4), confirming the presence of “theme gap" and “concept drift" challenges previously identified by Verberne et al. [9]. Though with this different experimental set-up, there is no need to worry for a Maximum recall which was a limitation in the work previously done by Verberne et al. The strategy of splitting texts into 384-token chunks and using max pooling across chunk predictions was crucial for handling long documents. Max pooling likely contributed to higher recall by predicting a label if any chunk exceeded the threshold. While effective, this method may inflate recall, suggesting future work could explore alternatives, such as averaging label predicting in chunks, to better estimate document-level themes.

7 Conclusion

This study examined multi-label classification of Dutch election manifestos in a small dataset with considerable label sparsity. Translation had minimal impact on performance: free OPUS-MT and commercial DeepL achieved nearly identical Micro-F1 scores, validating the chosen translation model for this task. BERT pre-training helped small models, with BERTje slightly outperforming base BERT on 380 labels, but model scale was the dominant factor. Where there is no bigger BERTje, Large BERT (3 times the parameter count) on translated text achieved Micro-F1 considerably larger than the baseline SVM, Base BERT and BERTje, showing that increasing model capacity provides substantial gains and outweighs the benefits of smaller, language-specific models.

These findings suggest that leveraging large English models via translation is an effective strategy for this multi-label classification task. The ap-

proach also improved recall on this task compared to traditional Linear SVMs. Future work should explore alternatives to chunking, optimal threshold for training, different pooling techniques when combining chunks, exploring different transformer models, translation models, or exploring different transformer settings.

8 AI Assistance

We have utilized ChatGPT to generate templates for LaTeX-tables, to rewrite sentences to be as concise and precise as possible, and written code to simplify data analysis for results that appear in this report.

Contributions

This work included translation, coding and experimentation(Ivor & Ivar), and writing(Ivor & Ivar).

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A All different threshold data

Table 6: BERTje performance on Dutch data

t	Included labels								
	First			First two			All three		
	F1	P	R	F1	P	R	F1	P	R
0.05	0.277	0.164	0.879	0.252	0.150	0.797	0.213	0.124	0.762
0.10	0.407	0.277	0.768	0.373	0.263	0.643	0.323	0.225	0.577
0.15	0.467	0.358	0.673	0.415	0.343	0.525	0.369	0.306	0.464
0.20	0.498	0.427	0.596	0.430	0.414	0.448	0.381	0.377	0.385
0.25	0.509	0.489	0.531	0.421	0.473	0.379	0.374	0.439	0.325
0.30	0.507	0.543	0.475	0.400	0.522	0.324	0.347	0.486	0.270
0.35	0.493	0.589	0.423	0.381	0.584	0.283	0.318	0.543	0.225
0.40	0.477	0.635	0.382	0.356	0.636	0.247	0.295	0.607	0.194
0.45	0.456	0.678	0.343	0.325	0.698	0.212	0.252	0.655	0.156
0.50	0.422	0.708	0.301	0.289	0.730	0.180	0.227	0.696	0.136

Table 7: BERT performance on English data

t	Included labels								
	First			First two			All three		
	F1	P	R	F1	P	R	F1	P	R
0.05	0.298	0.180	0.866	0.275	0.166	0.801	0.1870	0.1091	0.6541
0.10	0.406	0.278	0.754	0.371	0.259	0.651	0.2890	0.2130	0.4489
0.15	0.459	0.350	0.667	0.416	0.333	0.553	0.3262	0.3083	0.3464
0.20	0.497	0.417	0.614	0.434	0.397	0.479	0.3234	0.3826	0.2800
0.25	0.508	0.467	0.556	0.440	0.455	0.425	0.2954	0.4503	0.2198
0.30	0.510	0.514	0.505	0.432	0.505	0.378	0.2694	0.5174	0.1821
0.35	0.506	0.566	0.458	0.412	0.543	0.331	0.2314	0.5814	0.1445
0.40	0.493	0.609	0.414	0.386	0.588	0.288	0.1964	0.6474	0.1158
0.45	0.475	0.649	0.375	0.360	0.642	0.250	0.1641	0.7138	0.0927
0.50	0.456	0.684	0.341	0.327	0.669	0.216	0.1233	0.7333	0.0673

Table 8: Large BERT performance on English data

t	Included labels								
	First			First two			All three		
	F1	P	R	F1	P	R	F1	P	R
0.05	0.345	0.214	0.887	0.305	0.186	0.831	0.288	0.174	0.825
0.10	0.449	0.312	0.799	0.412	0.289	0.719	0.387	0.269	0.685
0.15	0.507	0.386	0.739	0.460	0.361	0.636	0.439	0.345	0.602
0.20	0.532	0.438	0.678	0.485	0.422	0.571	0.461	0.407	0.530
0.25	0.547	0.483	0.630	0.494	0.475	0.514	0.464	0.459	0.470
0.30	0.557	0.525	0.593	0.494	0.525	0.467	0.467	0.518	0.425
0.35	0.565	0.573	0.558	0.484	0.567	0.423	0.461	0.572	0.386
0.40	0.558	0.604	0.519	0.464	0.606	0.375	0.433	0.599	0.339
0.45	0.550	0.646	0.478	0.447	0.653	0.339	0.414	0.632	0.308
0.50	0.527	0.672	0.434	0.420	0.684	0.303	0.389	0.669	0.274